

1 2D-DCT Transform

(a)

For $A^{4 \times 4}$, $N=4$. Therefore, the DCT matrix T should be:

$$T[i, j] = \begin{cases} \sqrt{\frac{1}{N}} = \frac{1}{2}, & \text{if } i = 0 \\ \sqrt{\frac{2}{N}} \cos\left(\frac{(2j+1)i\pi}{2N}\right) = \frac{1}{\sqrt{2}} \cos\left(\frac{(2j+1)i\pi}{8}\right), & \text{if } 0 < i < N, 0 \leq j < N \end{cases}$$

$$T = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} \cos \frac{\pi}{8} & \frac{1}{2} \cos \frac{3\pi}{8} & \frac{1}{2} \cos \frac{5\pi}{8} & \frac{1}{2} \cos \frac{7\pi}{8} \\ \frac{1}{2} \cos \frac{2\pi}{8} & \frac{1}{2} \cos \frac{6\pi}{8} & \frac{1}{2} \cos \frac{10\pi}{8} & \frac{1}{2} \cos \frac{14\pi}{8} \\ \frac{1}{2} \cos \frac{3\pi}{8} & \frac{1}{2} \cos \frac{9\pi}{8} & \frac{1}{2} \cos \frac{15\pi}{8} & \frac{1}{2} \cos \frac{21\pi}{8} \end{bmatrix} \approx \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0.653281 & 0.270598 & -0.270598 & -0.653281 \\ 0.5 & -0.5 & -0.5 & 0.5 \\ 0.270598 & -0.653281 & 0.653281 & -0.270598 \end{bmatrix}.$$

As for the DCT transform consequence $D = TAT^T$

$$\text{Therefore, the consequence } D = \begin{bmatrix} 10.0000 & 9.2388 & 0.0000 & -3.8268 \\ 9.2388 & 8.5355 & 0.0000 & -3.5355 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ -3.8268 & -3.5355 & 0.0000 & 1.4645 \end{bmatrix}.$$

(b)

Comparing with the 2D-DFT, the 2D-DCT is easier to compute. The DCT only consider the cosine computation. The DFT must consider cosine and sine computation at same time. They will have similar performance on showing the color frequency on images. However, the DCT could save half of computation cost. Therefore, DCT performs better from my perspective.

2 Controllable Video Generation

Part 1:

- **Method Description:** Cross-subject motion-guided video generation aims to transfer human motion dynamics into realistic animal behaviors through pose-guided diffusion models.
- **Objective:** The objective is to generate biologically consistent videos where animals perform human-like actions naturally while preserving each species' motion traits.
- **Motivation:** This method would reduce the production cost of animal films and animations, enable controllable motion synthesis across species, and enhance visual realism and creativity in AI-assisted video generation. (95 Words)

Part 2:

References

- [1] Chan, C. et al., “Everybody Dance Now”, CVPR 2019.
- [2] Siarohin, A. et al., “First Order Motion Model for Image Animation”, NeurIPS 2019.
- [3] Zhang, X. et al., “Pose-Guided Person Image Generation”, ICCV 2019.